TEXORA · LECTURE SHEET

Light and Color

Colour begins before the eye: it is the visible slice of the electromagnetic spectrum, shaped by the illuminant and by what a material absorbs, reflects and transmits.

WP-4103 · COLOR PHYSICS · TOPIC 1 · SPECIALIZATION — WET PROCESSING

1. Electromagnetic radiation and the spectrum

Light is electromagnetic radiation — oscillating electric and magnetic fields travelling through space at the speed of light, $c = 3.00 \times 10^8$ m/s in vacuum. Radiation is described by its **wavelength** λ (distance between successive wave crests, in metres or nanometres, 1 nm = 10^{-9} m) and its **frequency** ν (oscillations per second, in hertz), linked by a fixed relation:

$$c = \lambda\nu \quad | \quad \nu = c / \lambda$$

The **electromagnetic spectrum** is the complete ordered range of this radiation. From long wavelength (low frequency, low energy) to short: radio waves, microwaves, infrared, **visible light**, ultraviolet, X-rays and gamma rays. The neighbouring regions matter in textiles: infrared is felt as heat and drives drying and heat-setting; ultraviolet causes photodegradation, yellowing of whites and fading of dyes, and is absorbed by UV-protective finishes; visible light does the seeing.

REGION	APPROX. WAVELENGTH	RELEVANCE TO TEXTILES
Ultraviolet (UV)	100–380 nm	Fading, fibre degradation, fluorescence of optical brighteners
Visible	**380–780 nm**	Colour perception and colour measurement
Infrared (IR)	780 nm – 1 mm	Heating, drying, curing; near-IR sorting of fibres

2. The visible region, 380–780 nm

The human visual system responds only to roughly **380–780 nm**. Within this narrow band, wavelength maps to hue: short wavelengths appear violet and blue, middle wavelengths green and yellow, long wavelengths orange and red. The boundaries are not sharp walls — sensitivity falls gradually at both ends — but 380–780 nm is the working range adopted in textile colour measurement, and

spectrophotometers for colour work normally scan this region (often 360–750 or 400–700 nm) in 10 or 20 nm steps.

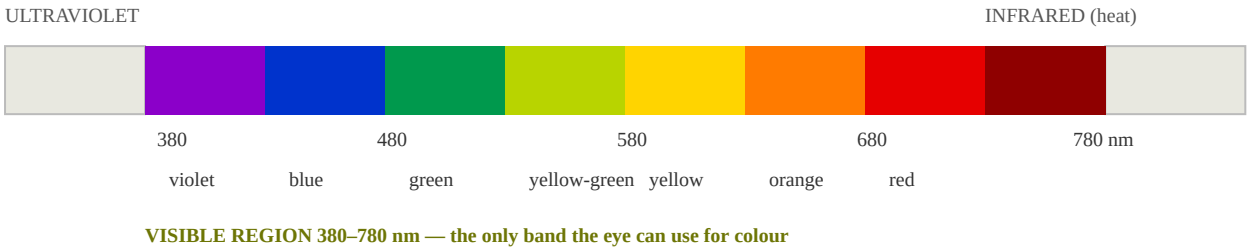


Figure 1 — The visible spectrum within the electromagnetic spectrum, with approximate hue bands and wavelength scale.

A beam containing all visible wavelengths in balanced proportions appears colourless — white. Sunlight, a tungsten lamp and a fluorescent tube all look “white” to an adapted eye, yet their wavelength compositions differ greatly (Section 6) — one root cause of shade disputes between dyehouse and buyer.

3. Light as photons — energy depends on wavelength

Light also behaves as a stream of particles called **photons**. Each photon carries an energy fixed by its frequency (Planck’s relation, $h = 6.626 \times 10^{-34}$ J·s):

$$E = h\nu = hc / \lambda$$

Short-wavelength photons are therefore more energetic than long-wavelength ones. This is why ultraviolet photons can break chemical bonds in dyes and fibres (fading, tendering), while infrared photons mainly excite molecular vibration — heat.

WORKED EXAMPLE 1 — PHOTON ENERGY AT THREE WAVELENGTHS

Using $c = 3.00 \times 10^{8}$ m/s, $h = 6.626 \times 10^{-34}$ J·s and 1 eV = 1.602×10^{-19} J:

450 nm (blue): $\nu = 3.00 \times 10^{8} / 450 \times 10^{-9} =$ **6.67×10^{14} Hz**; $E = 6.626 \times 10^{-34} \times 6.67 \times 10^{14} =$ **4.42×10^{-19} J = 2.76 eV**.

550 nm (green): $\nu =$ **5.45×10^{14} Hz**; $E =$ **3.61×10^{-19} J = 2.26 eV**.

650 nm (red): $\nu =$ **4.62×10^{14} Hz**; $E =$ **3.06×10^{-19} J = 1.91 eV**.

Check the pattern: halving the wavelength would double the energy — energy rises as wavelength falls.

4. How objects get their colour

An object does not “have” a colour in isolation; it modifies the light that reaches it. When light strikes a material, three things can happen at each wavelength:

- **Reflection** — light bounces back. On dyed textiles this is mostly diffuse reflection from within the fibre assembly; the reflected spectrum is what a spectrophotometer measures.
- **Absorption** — light energy is taken up, usually converted to heat. Dye molecules absorb selectively because their conjugated electronic structures accept photons of particular energies.
- **Transmission** — light passes through (transparent and translucent materials, thin fabrics against light).

For an opaque textile the three reduce to two: at each wavelength, reflectance $R(\lambda)$ + absorptance = 1. **Selective absorption** creates colour: a red fabric absorbs most blue and green light and reflects mainly long wavelengths, so the eye reports red. A **black** fabric absorbs strongly across the whole visible band; a **white** fabric reflects most light at all wavelengths with little selectivity; a **grey** reflects a roughly constant, smaller fraction. The **spectral reflectance curve** $R(\lambda)$ is thus the physical fingerprint of a shade — the starting point of all instrumental colour measurement (Topic 2).

APPEARANCE	SELECTIVE BEHAVIOUR	REFLECTANCE CURVE
White	Reflects most light, all wavelengths	High and flat
Black	Absorbs most light, all wavelengths	Low and flat
Red	Absorbs blue–green, reflects red	Low 400–600 nm, high 620–780 nm
Yellow	Absorbs blue, reflects green + red	Low 400–480 nm, high beyond 500 nm
Green	Absorbs blue and red, reflects green	Peak near 500–560 nm

5. Mixing colours — additive and subtractive

Two mixing systems operate in colour work, and they start from opposite ends.

Additive mixing mixes *lights*. The primaries are red, green and blue (RGB). Superposing lights adds their spectra, so mixtures are brighter than their components: red + green light = yellow, green + blue = cyan, red + blue = magenta, and all three in balance = white. Additive mixing governs displays, camera sensors and the DigiEye system (Topic 4).

Subtractive mixing mixes *colorants* — dyes, pigments, filters. Each colorant subtracts (absorbs) part of the spectrum, so mixtures are darker. The subtractive primaries are cyan, magenta and yellow (CMY), each absorbing its additive complement: cyan absorbs red, magenta absorbs green, yellow absorbs blue. Dyeing is subtractive mixing on a fibre: the classic trichromatic dye set (yellow, red, blue

— commercial equivalents of the CMY logic) builds a shade by jointly absorbing across the spectrum, which is why over-dyeing can only deepen or dull a shade, never lighten it.

FEATURE	ADDITIVE	SUBTRACTIVE
What is mixed	Coloured lights	Dyes, pigments, filters
Primaries	Red, green, blue	Cyan, magenta, yellow
All primaries together	White	Black (in practice, a dark muddy shade)
Brightness of mixture	Increases	Decreases
Where used	Screens, cameras, DigiEye	Dyeing, printing, paints

6. Illuminants and spectral power distribution

An **illuminant** is a light source defined (or realised) by its **spectral power distribution (SPD)** — the relative power it emits at each wavelength. The SPD multiplies the fabric's reflectance in the eye and in every colour calculation, so the same fabric genuinely presents a different stimulus under different illuminants.

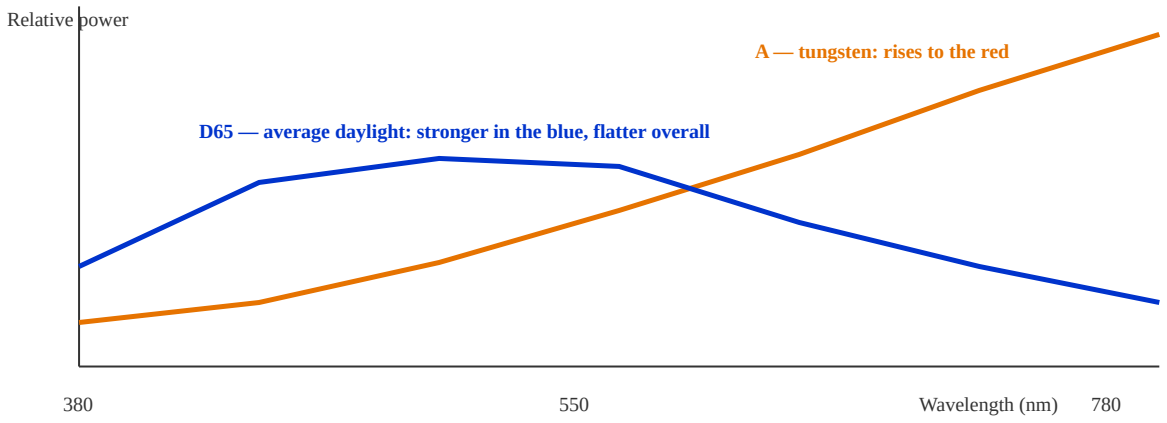


Figure 2 — Sketch of spectral power distributions: illuminant A (tungsten) versus D65 (average daylight). Curves are schematic shapes, normalised for comparison.

The CIE defines **standard illuminants** as published SPD tables so that laboratories everywhere calculate with identical "light":

ILLUMINANT	REPRESENTS	APPROX. COLOUR TEMPERATURE	TEXTILE USE
D65	Average daylight, including UV region	6500 K	Primary illuminant for shade sorting and instrumental pass/fail

A	Tungsten filament (incandescent) lamp	2856 K	Metamerism check — warm domestic light
F11 / TL84	Narrow-band fluorescent retail/store lighting	4000 K	Metamerism check — shop lighting specified by many retailers
CWF (F2)	Cool-white fluorescent, office lighting	4150 K	Secondary check for some markets

A physical **light cabinet** houses real lamps approximating these illuminants so visual assessment matches the instrumental conditions. A shade approved only under D65 may fail in store light — the phenomenon of **illuminant metamerism** (Topic 3) — which is why professional shade approval is a multi-illuminant judgement.

7. Colour temperature

Correlated colour temperature (CCT), in kelvin (K), describes the warmth or coolness of a near-white source by comparison with a heated black-body radiator. Low CCT (≈ 2800 K) is warm and reddish — tungsten light; 4000 K is neutral; D65 at ≈ 6500 K is a cooler, bluer daylight. Colour temperature is a one-number summary only: two sources with the same CCT can have different SPDs (smooth daylight versus spiky fluorescent) and render colours differently. For colour-critical work the full SPD is what counts, with CCT as a convenient label.

8. Glossary — key terms

- **Wavelength (λ):** distance between wave crests; colour-relevant unit is the nanometre.
- **Photon:** quantum of light with energy $E = hc/\lambda$.
- **Spectral reflectance curve:** reflectance of a sample at each wavelength — the physical fingerprint of a shade.
- **Illuminant:** a light source specified by its spectral power distribution.
- **Spectral power distribution (SPD):** relative emitted power versus wavelength.
- **Correlated colour temperature:** warmth/coolness of a source in kelvin, referenced to a black-body radiator.
- **Additive / subtractive mixing:** mixing lights (RGB, brighter) versus mixing colorants (CMY, darker).

9. Key points

- Visible light is only 380–780 nm of the electromagnetic spectrum; photon energy rises as wavelength falls.
- Object colour = illuminant SPD × selective absorption/reflection/transmission of the material.
- Dyeing is subtractive mixing; displays and cameras are additive.
- Standard illuminants D65 (primary), A and F11/TL84 (checks) make shade approval reproducible.

10. Summary

Colour physics starts with radiation: a narrow visible band, photons whose energy is set by wavelength, and materials that sculpt that band by selective absorption and reflection. Add the illuminant, described by its SPD, and the stimulus reaching the eye is fixed. Colorimetry (Topic 2) turns it into numbers; human vision (Topic 3) explains how they are perceived.

11. Practice

1. Calculate the frequency and photon energy (J and eV) of light of wavelength 500 nm.
2. A fabric reflects 75% at 650 nm and 10% at 450 nm. Describe its likely appearance and explain it in terms of selective absorption.
3. Why can two lamps of the same colour temperature render a shade differently?
4. A dyer adds more dye to correct a batch that is too light, and the shade also becomes duller. Explain using subtractive mixing.
5. State the primary illuminant for textile shade approval and two illuminants used as metamerism checks.

Answer key

1. $\nu = 3.00 \times 10^{8} / 500 \times 10^{-9} = 6.00 \times 10^{14}$ Hz. $E = 6.626 \times 10^{-34} \times 6.00 \times 10^{14} = 3.98 \times 10^{-19}$ J = $3.98 \times 10^{-19} / 1.602 \times 10^{-19} = 2.48$ eV.
2. Reddish: it strongly reflects long (red) wavelengths while absorbing most short (blue) light, so the reflected stimulus is dominated by red.
3. CCT is a single summary number; the full SPDs can differ (smooth versus spiky fluorescent spectra), and colour rendering depends on the whole SPD multiplied by the sample's reflectance.
4. Each added dye absorbs more of the spectrum (subtractive). Total reflectance falls — deeper — and because absorption broadens across wavelengths, chroma falls too — duller.
5. Primary: D65. Checks: illuminant A (tungsten) and F11/TL84 (store fluorescent).